



SVKM'S NMIMS
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING/
SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2023-2024

Program/s: B Tech

Year: II Semester: IV

Stream/s ; AI

Subject: Machine Learning

Time: 3 hrs (10 am to 1 pm)

Date: 18 / 04 / 2024

No. of Pages: 4

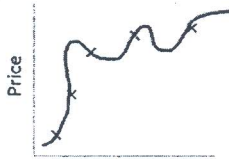
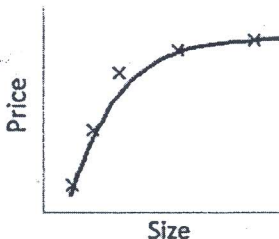
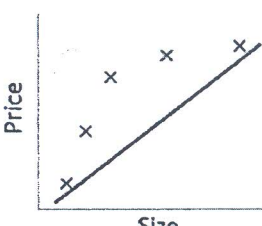
Marks: 100

Final Examination/Re-Examination (2022-23)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) **In all 5 questions to be attempted.**
- 4) All questions carry equal marks.
- 5) **Answer to each new question to be started on a fresh page.**
- 6) **Figures in brackets on the right hand side indicate full marks.**
- 7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	
CO3 ; SO-6 ; BL-3	a.	Suppose you developed a model to predict whether a received email is spam (positive class) or not spam (negative class). You tested the model on 100 emails and the results were: - True Positives (TP): 40 emails - True Negatives (TN): 50 emails - False Positives (FP): 5 emails - False Negatives (FN): 5 emails Calculate the Precision, Recall, Accuracy and F1 score. Also, show the confusion matrix.	[5]
CO1 ; SO-6 ; BL-2	b.	Describe the scenario where machine learning is boon for the society. Identify the types of machine learning algorithms. In case of supervised algorithm, also specify the types. Justify your answers i. Stock market prediction ii. Prediction of count of potential dengue patients iii. Placing of items in shopping mart	[5]
CO2- ; SO- 1; BL-3	c.	Differentiate between Lasso and Ridge Regression. Use schematics.	[5]

CO3- ; SO- 6; BL-4	d.	For the below feature space (Euclidian distance given) for 5 points, identify the core/border/noise points for DBSCAN given EPS=0.2 and MinPts=2. <table><tr><td></td><td>P1</td><td>P2</td><td>P3</td><td>P4</td><td>P5</td></tr><tr><td>P1</td><td>0</td><td>0.9</td><td>0.59</td><td>0.45</td><td>0.65</td></tr><tr><td>P2</td><td>0.9</td><td>0</td><td>0.36</td><td>0.53</td><td>0.02</td></tr><tr><td>P3</td><td>0.59</td><td>0.36</td><td>0</td><td>0.56</td><td>0.15</td></tr><tr><td>P4</td><td>0.45</td><td>0.53</td><td>0.56</td><td>0</td><td>0.24</td></tr><tr><td>P5</td><td>0.65</td><td>0.02</td><td>0.15</td><td>0.24</td><td>0</td></tr></table>		P1	P2	P3	P4	P5	P1	0	0.9	0.59	0.45	0.65	P2	0.9	0	0.36	0.53	0.02	P3	0.59	0.36	0	0.56	0.15	P4	0.45	0.53	0.56	0	0.24	P5	0.65	0.02	0.15	0.24	0	[5]
	P1	P2	P3	P4	P5																																		
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Q2 a CO1-; SO-1; BL-4		i. Determine the underfitted and overfitted model from the graphs given below. Justify your answer. Discuss problem of underfitting and overfitting while designing machine learning model. How can you overcome it?  $\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$ (a)  $\theta_0 + \theta_1 x + \theta_2 x^2$ (b)  $\theta_0 + \theta_1 x$ (c) ii. Using Apriori algorithm, find the frequent item sets for the following transactions and items (Minimum Support = 2) <table><tr><td>Transaction</td><td colspan="4">Items</td></tr><tr><td>T1</td><td>1</td><td>3</td><td>4</td><td></td></tr><tr><td>T2</td><td>2</td><td>3</td><td>5</td><td></td></tr><tr><td>T3</td><td>1</td><td>2</td><td>3</td><td>5</td></tr></table>	Transaction	Items				T1	1	3	4		T2	2	3	5		T3	1	2	3	5	[10]																
Transaction	Items																																						
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Q2 b CO2-; SO-1; BL-5		Demonstrate using diagram the concept of support vectors in SVM. Show that in SVM, minimizing the weights maximizes the margin. Explain the importance of soft margin in SVM. How does SVM deal with nonlinear data?	[10]																																				
Q3 a CO3-; SO-1; BL-5		Apply the concept of Naïve bayes classifier for text classification to classify: <i>A very close game</i> <table><tr><td>Text</td><td>Tag</td></tr><tr><td>“A great game”</td><td>Sports</td></tr><tr><td>“The election was over”</td><td>Not sports</td></tr><tr><td>“Very clean match”</td><td>Sports</td></tr><tr><td>“A clean but forgettable game”</td><td>Sports</td></tr><tr><td>“It was a close election”</td><td>Not sports</td></tr></table>	Text	Tag	“A great game”	Sports	“The election was over”	Not sports	“Very clean match”	Sports	“A clean but forgettable game”	Sports	“It was a close election”	Not sports	[10]																								
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Q3 b CO2-; SO-1; BL-5	Determine values of thetas by performing two iterations of gradient descent for linear regression. (use vectorized method). Let learning rate be 0.001, start with initial coefficients as 0. Calculate the error at each iteration. <table><tr><th>X</th><th>Y</th></tr><tr><td>1</td><td>8</td></tr><tr><td>2</td><td>17</td></tr><tr><td>3</td><td>10</td></tr><tr><td>3</td><td>22</td></tr><tr><td>4</td><td>20</td></tr><tr><td>4</td><td>27</td></tr><tr><td>5</td><td>50</td></tr></table>	X	Y	1	8	2	17	3	10	3	22	4	20	4	27	5	50	[10]					
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5	50																						
Q4a CO3-; SO-6; BL-4	i. Describe with schematics how Logistic regression can be used for multi-class classification. [5] ii. Consider the below dataset: <table><tr><th>X1</th><th>X2</th><th>Default</th></tr><tr><td>87</td><td>2</td><td>N</td></tr><tr><td>89</td><td>2</td><td>N</td></tr><tr><td>100</td><td>26</td><td>Y</td></tr><tr><td>275</td><td>54</td><td>Y</td></tr><tr><td>42</td><td>57</td><td>Y</td></tr><tr><td>88</td><td>53</td><td>N</td></tr></table> The data set was used as training set to build a logistic regression model and the values of thetas obtained after training were $\theta_0 = -5.7$, $\theta_1 = 0.008$ and $\theta_2 = 0.102$. Consider the data you want to test on the model is the training set (above). Find the new predictions. [5]	X1	X2	Default	87	2	N	89	2	N	100	26	Y	275	54	Y	42	57	Y	88	53	N	[10]
X1	X2	Default																					
87	2	N																					
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Q4 b CO3-; SO-1; BL-5	Apply 2 iterations of K-means clustering and get the centroids (K=3) on the following data samples. Assume initial centroids: C1=(2,3), C2=(5,4),C3=(9,6). Calculate Cluster Cohesion (SSE) and Cluster Separation (SSB). <table><tr><th>x1</th><th>x2</th></tr><tr><td>2</td><td>3</td></tr><tr><td>5</td><td>4</td></tr><tr><td>9</td><td>6</td></tr><tr><td>8</td><td>2</td></tr><tr><td>7</td><td>5</td></tr><tr><td>6</td><td>8</td></tr><tr><td>3</td><td>6</td></tr><tr><td>4</td><td>2</td></tr></table>	x1	x2	2	3	5	4	9	6	8	2	7	5	6	8	3	6	4	2	[10]			
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Q5 a CO1; SO1; BL 5		Apply agglomerative clustering using single linkage approach on the following data set. Plot the dendrogram of your result. A(1, 1), B(2, 3), C(3, 5), D(4,5), E(6,6), and F(7,5)	[10]																																	
Q5 b CO3-; SO6; BL-4		What is Ensemble learning? What is the difference between Bagging and Boosting techniques?	[10]																																	
Q6 a CO3-; SO1; BL-5		Demonstrate Principal component analysis to perform dimensionality reduction on the following data points: (4,11), (8,4) (13,5) (7,14).	[10]																																	
Q6 b CO1-; SO-1; BL- 4		i. Describe the major tasks required in data preprocessing. [5] ii. Explain the different methods of choosing the test set for model evaluation in machine learning, which is the best and why? [5]	[10]																																	
Q7 CO3-; SO1; BL-6		Consider the following dataset for a binary class problem: <table><thead><tr><th>A</th><th>B</th><th>Class Label</th></tr></thead><tbody><tr><td>T</td><td>F</td><td>+</td></tr><tr><td>T</td><td>T</td><td>+</td></tr><tr><td>T</td><td>T</td><td>+</td></tr><tr><td>T</td><td>F</td><td>-</td></tr><tr><td>T</td><td>T</td><td>+</td></tr><tr><td>F</td><td>F</td><td>-</td></tr><tr><td>F</td><td>F</td><td>-</td></tr><tr><td>F</td><td>F</td><td>-</td></tr><tr><td>T</td><td>T</td><td>-</td></tr><tr><td>T</td><td>F</td><td>-</td></tr></tbody></table> i. Calculate the information gain when splitting on A and B. Which attribute would the decision tree induction algorithm choose (use Entropy as an impurity measure)? ii. Calculate the gain in the Gini index when splitting on A and B. Which attribute would the decision tree induction algorithm choose?	A	B	Class Label	T	F	+	T	T	+	T	T	+	T	F	-	T	T	+	F	F	-	F	F	-	F	F	-	T	T	-	T	F	-	[20]
A	B	Class Label																																		
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